

targettype energy rate_10_11C num_10_11C num_12C all_carbon					
0	c_1086	400	0.77	3746.66	485536.34 489283
1	c_1086	550	0.86	3426.20	394401.80 397828
2	c_1086	650	0.78	3597.89	454955.11 458553
3	c_1086	800	0.66	1703.97	255458.03 257162
4	c_2198	400	1.28	6373.54	490387.46 496761
5	c_2198	550	1.50	5537.30	362626.70 368164
6	c_2198	650	1.37	5999.79	432899.21 438899
7	c_2198	800	1.27	4089.52	318206.48 322296
8	c_54	400	0.49	2305.75	470899.25 473205
9	c_54	550	0.49	2175.51	440189.49 442365
10	c_54	650	0.42	2063.60	492394.40 494458
11	c_54	800	0.43	1263.02	295726.98 296990
12	empty	400	0.15	748.23	495474.77 496223
13	empty	550	0.21	960.83	451309.17 452270
14	empty	650	0.14	732.45	513272.55 514005
15	empty	800	0.15	464.19	316850.81 317315

```
#x params...
mean_x_thin = -2.713415
sigma_x_thin = 3.400895

mean_x_medium = -2.673165
sigma_x_medium = 3.7922116

mean_x_thick = -2.71375
sigma_x_thick = 4.537883

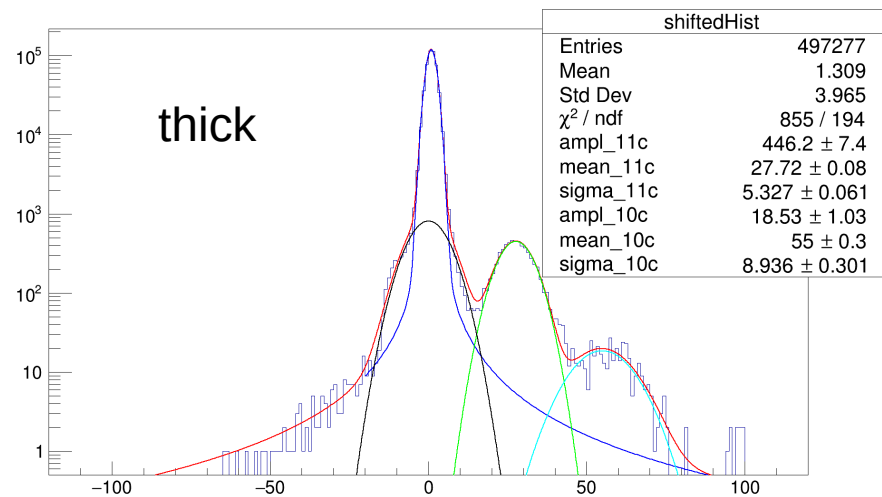
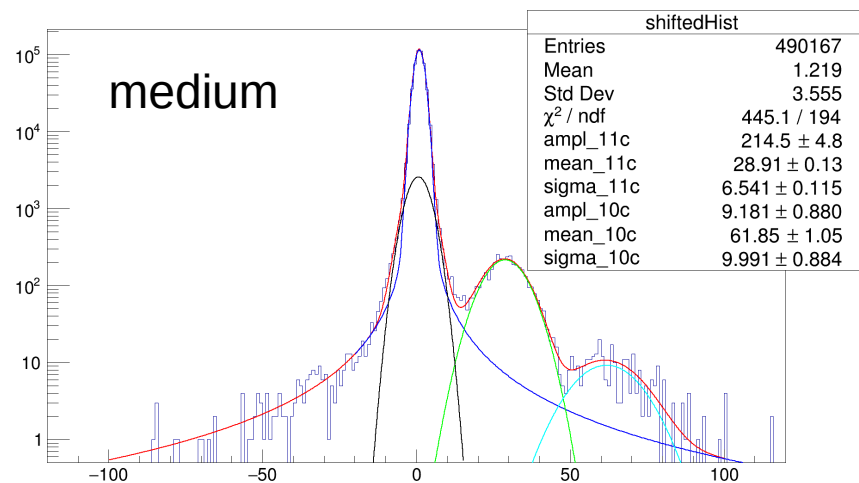
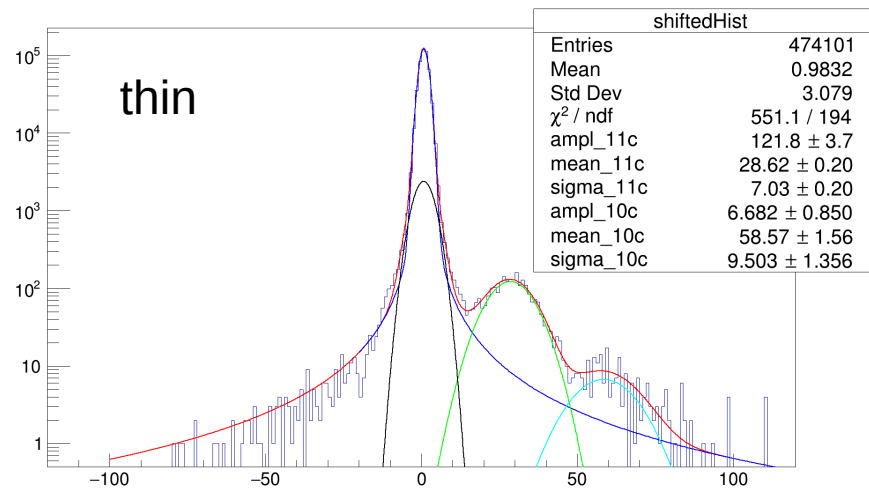
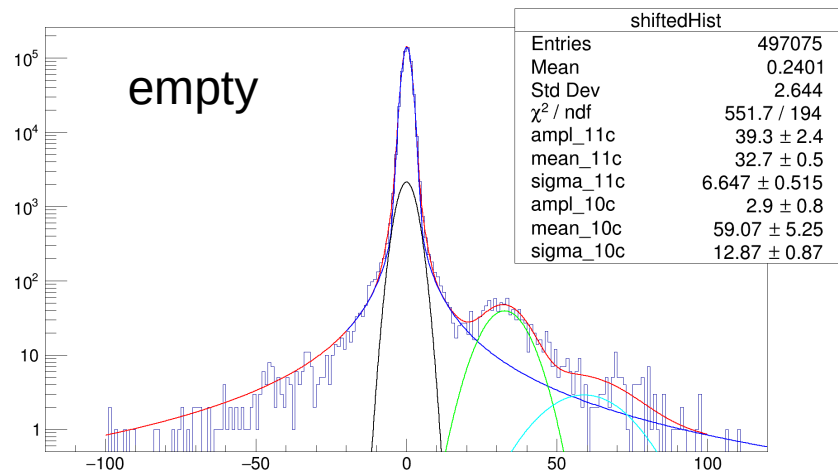
mean_x_empty = -2.647455
sigma_x_empty = 2.9543483

#y params...
mean_y_thin = 14.9058805
sigma_y_thin = 5.1731065

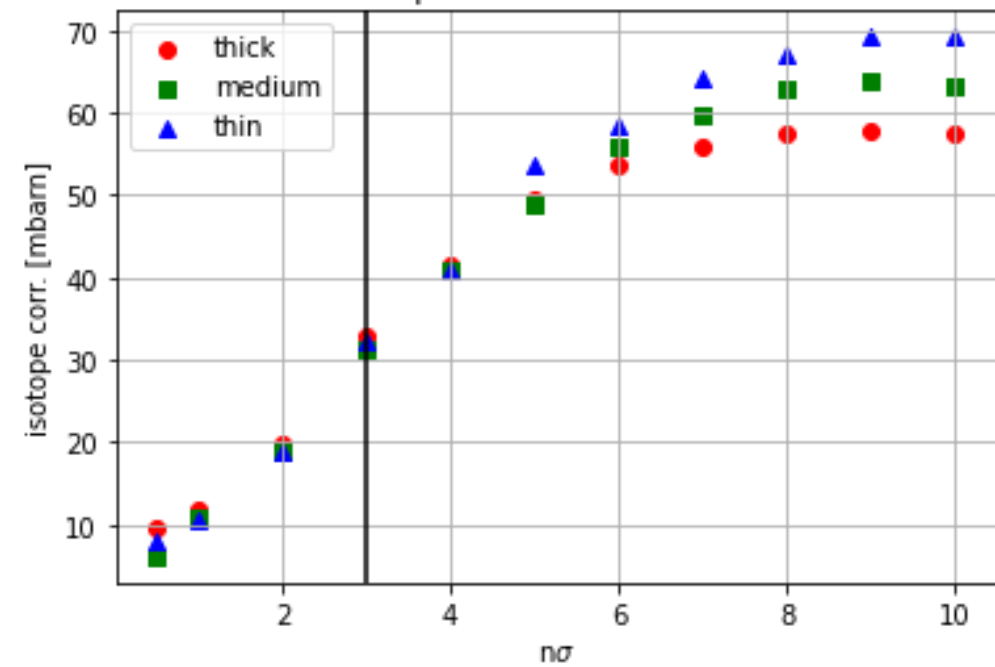
mean_y_medium = 14.811775
sigma_y_medium = 5.8811083

mean_y_thick = 14.67804
sigma_y_thick = 7.2668183

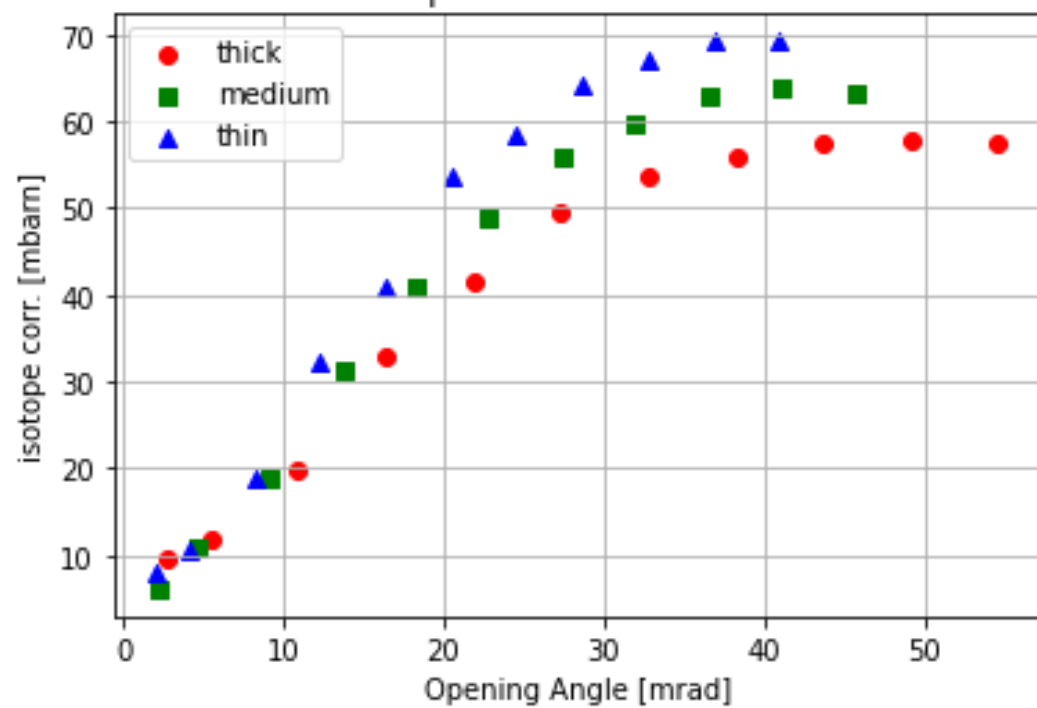
mean_y_empty = 15.01774
sigma_y_empty = 4.39692
```



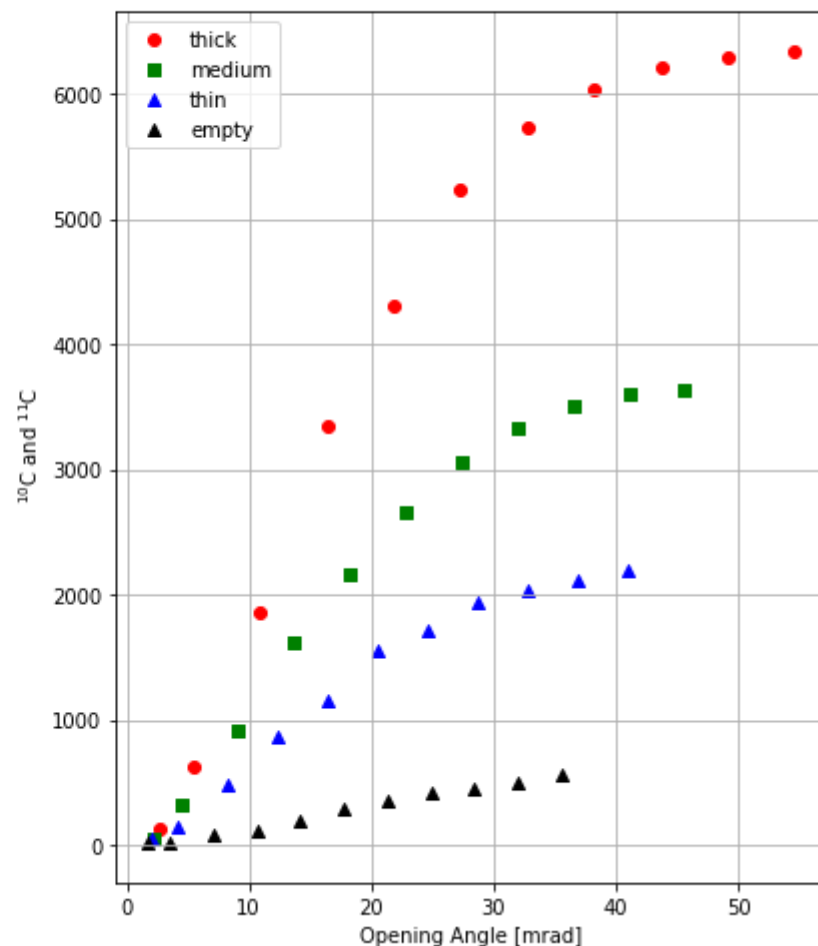
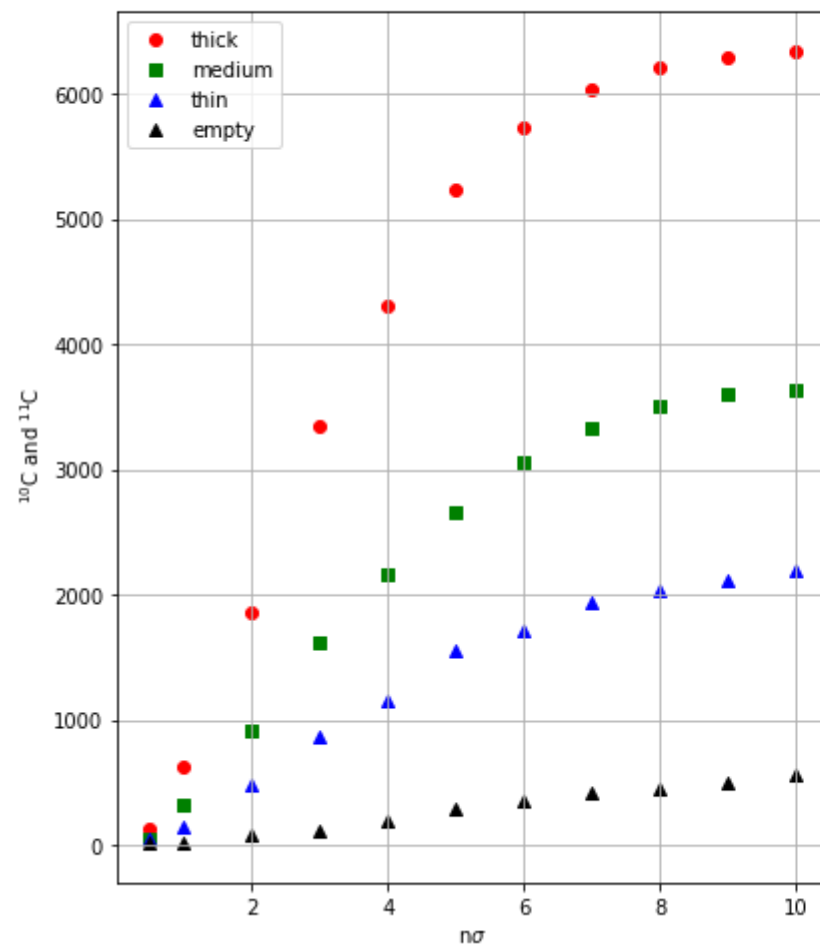
Isotope correction in mbarn



Isotope correction in mbarn



Reconstructed $^{10}\text{C}/^{11}\text{C}$



Reconstructed ^{12}C 